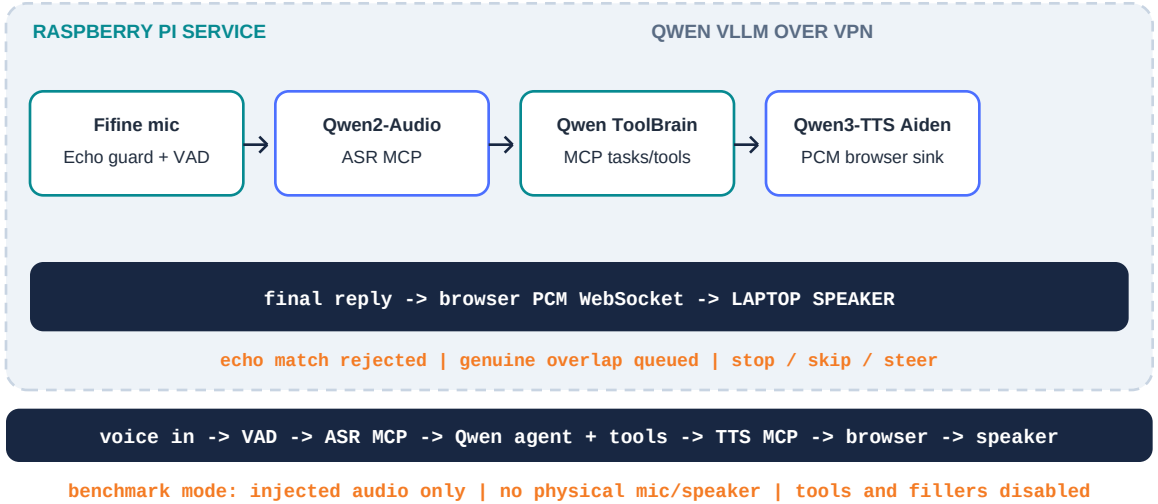
Full-Duplex Voice Agent on a Raspberry Pi

1 SYSTEM DIAGRAM

input -> Pi -> agent -> laptop speaker



- Fifine mic is captured by the Pi service; laptop browser is the speaker sink in the intended demo route.
- Pi owns echo guard, Silero VAD, MCP servers, task safety, and duplex state; Qwen services are reached over VPN.
- Qwen2-Audio ASR -> Qwen ToolBrain / MCP -> Qwen3-TTS Aiden; output is 16-kHz PCM to the browser.
- The agent supports web/file/build tools and direct voice commands; no model training is required.
- Protected overlap keeps listening open: matched echo is rejected; genuine speech is queued; stop, skip, and steer are explicit controls.

2 PROMPT

versioned in config/config.yaml

SYSTEM PROMPT (QWEN FAST MODE) - EXCERPT

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Filipino / English / Taglish voice assistant for a Philippines customer-support pilot.
Formal, respectful, concise spoken replies; never flirtatious or suggestive.
Execute clear user requests with bounded tools; ask only for missing details.
Never expose credentials or claim an external action succeeded without tool evidence.
For long work: "I am processing your request now."
```

3 CONTEXT ENGINEERING

short, safe, observable turns

GOES IN:

- system persona + PH language policy
- ASR transcript and VAD endpoint
- rolling history + tool results
- overlap, echo, and control events

MANAGED:

- fast mode; hidden thinking off
- 24-round tool budget + safe summary
- 1-3 spoken sentences; short history
- latest-prompt queue + background status
- sanitized terminal; no credentials or CoT

MCP Services, Evaluation & Model Serving

build + run owned MCP services on the RPi

4 MCP SERVICES

+ model serving: one table, one truth

Component	MCP server / transport	Tools	Model	Served where / notes
ASR	me3-asr / FastMCP	transcribe(audio) -> text	Qwen2-Audio-7B-Instruct	Pi service; benchmark used faster-whisper fallback
TTS	me3-tts / FastMCP	speak(text) -> audio	Qwen3-TTS-12Hz-1.7B-CustomVoice; Aiden	Pi service; Qwen vLLM over VPN; one formal voice
LLM	ToolBrain / MCP client	function calling	Qwen/Qwen3.8-27B	Remote vLLM; fast_mode=true; hidden thinking=false
VAD	me3-vad / FastMCP	detect_speech(frames)	Silero ONNX	On-device Pi CPU; 16 kHz mono; endpointing
Task + channels	me3-tasks / FastMCP + REST	files, web, build, timers, ME2 parity, channels	n/a / provider APIs	Pi-owned tools; lights/thermostat simulated; APIs need setup

5 EVALUATION & BENCHMARK

updated class direct-feed protocol

- One clock: benchmark direct feed uses the assistant machine clock and records stage marks per turn.
- Injection only: no physical microphone or speaker; no acoustic feedback in the latency run.
- Benchmark mode: no tool schemas, external agent/harness calls, fillers, backchannels, or real channel side effects.
- Report mean + p95 over the updated stage-latency phrases and preserve JSON, CSV, HTML, and latency chart artifacts.

Metric	Definition	Target	Mean	p95
E2E latency	end of speech -> audio out	4.0 s	5.81 s	7.87 s
ASR final	VAD end -> transcript	0.8 s	3.38 s	4.92 s
LLM TTFT	transcript -> first token	1.5 s	0.21 s	0.21 s
TTS TTFB	text -> first audio chunk	0.6 s	1.60 s	2.98 s
Barge-in	speech -> playback stopped	0.8 s	n/m	n/m
WER	ASR accuracy on test set	15%	29.3%	n/a

DEMO (deck 2)

Operator confirmed Pi Fifine speech -> ME3 -> laptop browser speaker.
Direct commands work without Qwen; complex tasks need model access.
Barge-in was not measured in the injection-only benchmark (n/m).

RESULTS (deck 2)

30 planned; 29 replies and one no-reply. All valid trials aggregated; no best run selected.
LLM TTFT met target; E2E, ASR, TTS, and WER missed.
ASR used faster-whisper: DGX3 port 8000 was unreachable.

CHALLENGES & REFERENCES

Restore DGX3 ASR; rerun the benchmark and measure barge-in.
Spotify needs API access, Premium, and PKCE; current account lacks API access.
Calendar OAuth, Telegram bot setup, and Android call permissions remain.
Lights/thermostat are simulated. See the benchmark report.